



ADAPTIVE EDGE ELEMENT METHOD FOR A QUASILINEAR PROBLEM IN ELECTROMAGNETISM WITH STRONG CONVERGENCE FOR GAUSS' LAW

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This paper is concerned with adaptive edge element methods based on a regularized formulation([1]-[3]) for a quasilinear $H(\text{curl})$ -elliptic problem from electromagnetism. The lowest order 1st-kind and 2nd-kind Nédélec elements are employed in discrete problems. Our theoretical results include an $H(\text{curl})$ -strong convergence of the sequence of discrete solutions by the proposed adaptive algorithm towards the exact solution and a vanishing limit of the sequence of relevant error estimators. Moreover, the adaptive algorithm is shown to ensure a strong convergence for the associated discrete Gauss' law in the H^{-1} -norm. Finally, with the Dörfler marking included, the adaptive algorithm is shown to be contractive for the sum of an $H(\text{curl})$ -error related quantity and a scaled estimator up to a high order term between two consecutive adaptive loops.

References

- [1] Ciarlet, P., Wu, H., & Zou, J. (2014). Edge Element Methods for Maxwell's Equations with Strong Convergence for Gauss' Laws. *SIAM Journal on Numerical Analysis*, 52(2), 779-807. DOI: 10.1137/120899856
- [2] Duan, H., Li, S., Tan, R. C., & Zheng, W. (2012). A delta-regularization finite element method for a double curl problem with divergence-free constraint. *SIAM Journal on Numerical Analysis*, 50(6), 3208-3230. DOI: 10.1137/110850578
- [3] Reitzinger, S. and Schöberl, J. (2002), An algebraic multigrid method for finite element discretizations with edge elements. *Numer. Linear Algebra Appl.*, 9: 223-238. DOI: 10.1002/nla.271